CCA Project

Yunus GÜRLEK

Frederic XIA

January 25, 2026

Contents

1 Resultant

For the entire section, let $f, g \in \mathbb{K}[x]$, where $\mathbb{K}$ is a field, with $\deg f = n$ and $\deg g = m$.

First, we define the Sylvester matrix, $S(f,g) \in \mathbb{Z}^{(n+m)\times(n+m)}$, as follows [1] :

$$S(f,g) = \begin{pmatrix} f_n & & & & g_m & & \\ f_{n-1} & f_n & & & g_{m-1} & g_m & \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots \\ f_0 & f_1 & \cdots & f_n & g_0 & g_1 & \cdots \\ & f_0 & \ddots & \vdots & & g_0 & \ddots \\ & & \ddots & f_1 & & & \ddots \\ & & & f_0 & & & g_0 \end{pmatrix}$$

where all the non-specified coefficients are null. We can observe that the first m columns of $S(f,g)$ are coefficients of $f(x)$ shifted m times, and the last n columns are coefficients of $g(x)$ shifted n times. We can also define the matrix with reverse order of f and g, or by shifting in rows instead of columns.

Then, the resultant of f and g is defined as [1]:

$$res(f,g) = \det S(f,g)$$

The resultant is important, as we have the following property:

$$res(f,g) \neq 0 \iff \gcd(f,g) = 1$$

which implies that the resultant of two polynomials is 0 if and only if they have a common root.

Proof. Let $\mathbb{K}[x]_{<d}$ be the $\mathbb{K}$-vector space with polynomials of degree $< d$, and let its standard bases $(1, x, \dots, x^{d-1})$.

First, let us reexamine the definition of the Sylvester matrix: We define the $\mathbb{K}$ linear map:

$$\phi : \mathbb{K}[x]_{<m} \oplus \mathbb{K}[x]_{<n} \to \mathbb{K}[x]_{<n+m}$$

such that:

$$\phi(u,v) = uf + vg$$

This definition is a linear map since the addition of polynomials and multiplication of polynomials with a constant polynomial is linear.

Then, we can observe that the matrix of ϕ in the standard bases is exactly the Sylvester matrix, $S(f,g)$. Hence,

$$res(f,g) = 0 \iff \det(S(f,g)) = 0 \iff \det(\phi) = 0 \iff \ker(\phi) \neq \emptyset$$

As a result, we can write:

$$res(f,g) = 0 \iff \exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0$$

We show the contraposition, $\gcd(f,g) \neq 1 \iff res(f,g) = 0$.

($\implies$): Assume $\gcd(f,g) \neq 1$. Then $\exists h \in \mathbb{K}[x], \deg(h) > 0 : h \mid f$ and $h \mid g$. Let $f_1, g_1 \in \mathbb{K}[x] : f = hf_1, g = hg_1$ and set $u = g_1, v = -f_1$. Then,

$$uf + vg = g_1 hf_1 - f_1 hg_1 = h \times (g_1 f_1 - f_1 g_1) = 0$$

As $\deg(h) > 0$, we have $\deg(u) < m, \deg(v) < n$. Thus, by the above equality,

$$\gcd(f,g) \neq 1 \implies \exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0 \implies res(f,g) = 0$$

($\Longleftarrow$): Assume $res(f,g) = 0$. Then, by the above equality we have

$$(\exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0) \implies uf = -vg$$

We know $u \neq 0$, as g is not 0 and u and v cannot be both 0. Let $d = \gcd(u,g)$ such that:

$$u = du_1, g = dg_1 \text{ where } gcd(u_1, g_1) = 1$$

Putting this on the above equation:

$$(du_1) \times f = -v \times (dg_1) \implies u_1 f = -vg_1$$

$$\implies g_1 \mid u_1 f \implies g_1 \mid f \text{ since } gcd(u_1, g_1) = 1$$

Thus, we now have g_1 as the common factor of g and f. Now showing that g_1 is not constant. We know $\deg(u) < m = \deg(g)$:

$$\deg(u) < \deg(g) \implies \deg(du_1) < \deg(dg_1)$$

$$\implies \deg(d) + \deg(u_1) < \deg(d) + \deg(g_1) \implies \deg(u_1) < \deg(g_1)$$

Since we have $\deg(u_1) \geq 0$, $\deg(g_1) > 0$. This shows that g_1 is a non-constant common divisor of f and g, so $\gcd(f,g) \neq 1$.

We conclude that

$$res(f,g) \neq 0 \iff \gcd(f,g) = 1$$

$\square$

2 Subresultants

As before, for the entire section, let $f, g \in \mathbb{Z}[x]$ with $\deg f = n$ and $\deg g = m$. Moreover, we assume $m \leq n$.

Resultant of polynomials is important, as it gives as an efficient manner to understand if two polynomials have a common root or not. However, it does not give any other information but about gcd. Thus, we can extend the resultant definition into subresultants, which include all results of the Extended Euclidean Algorithm [2].

First, we define Sylvester matrices for subresultants, similar to the resultant's matrix. Given $0 \leq k \leq m$,

$$S_k(f,g) = \begin{pmatrix}
f_n & & & & g_m & & & & \\
f_{n-1} & f_n & & & g_{m-1} & g_m & & & \\
\vdots & & \ddots & & \vdots & & \ddots & & \\
f_{n-m+k+1} & \cdots & \cdots & f_n & g_{k+1} & \cdots & \cdots & g_m & \\
\vdots & & & \vdots & \vdots & & & & \ddots \\
f_{k+1} & \cdots & \cdots & f_m & g_{m-n+k+1} & \cdots & \cdots & \cdots & \cdots & g_m \\
\vdots & & & \vdots & \vdots & & & & & \vdots \\
f_{2k-m+1} & \cdots & \cdots & f_k & g_{2k-n+1} & \cdots & \cdots & \cdots & \cdots & g_k
\end{pmatrix}$$

, which can be understand as removing the last k-columns of each submatrix of f and g, and then removing the last $2k$ rows of the matrix.

Then, the k-th subresultant of f and g is defined as [2]:

$$\alpha_k = \det S_k(f,g)$$

References

[1] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra.* Cambridge University Press, Cambridge, 1999. Section 6.3: The resultant.

[2] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra.* Cambridge University Press, Cambridge, 1999. Section 6.10: Subresultants.